

$k=1$

$$T(1) = 1 \quad \swarrow$$

$$T(n) = 3T\left(\frac{n}{4}\right) + n^2$$

$k=0$

$$3T\left(\frac{n}{4}\right) + n^2$$

$k=1$

$$3\left[3T\left(\frac{n}{4^2}\right) + \left(\frac{n}{4}\right)^2\right] + n^2$$

$$3^2 T\left(\frac{n}{4^2}\right) + 3\left(\frac{n}{4}\right)^2 + n^2$$

$k=2$

$$3^2 \left[3T\left(\frac{n}{4^3}\right) + \left(\frac{n}{4^2}\right)^2\right] + 3\left(\frac{n}{4}\right)^2 + n^2$$

$$3^3 T\left(\frac{n}{4^3}\right) + 3^2 \left(\frac{n}{4^2}\right)^2 + 3\left(\frac{n}{4}\right)^2 + n^2$$

$k=3$

$$3^3 \left[3T\left(\frac{n}{4^4}\right) + \left(\frac{n}{4^3}\right)^2\right] + 3^2 \left(\frac{n}{4^2}\right)^2 + 3\left(\frac{n}{4}\right)^2 + n^2$$

$$\sum \left(3^4 T\left(\frac{n}{4^4}\right) + 3^3 \left(\frac{n}{4^3}\right)^2 + 3^2 \left(\frac{n}{4^2}\right)^2 + 3 \left(\frac{n}{4}\right)^2 + n^2 \right)$$

$$K = N$$

$$3^K T\left(\frac{n}{4^K}\right) + \sum_{i=0}^{K-1} 3^i \left(\frac{n}{4^i}\right)^2 \quad \left(\frac{n^2}{16^i}\right)$$

$$\frac{n}{4^K} = 1 \rightarrow \log_4 n = K$$

$$4^K = n \rightarrow \log_4 n = K \quad K = \log_4 n$$

$$\boxed{a^b = c}$$

$$\log_a c = b$$

$$\rightarrow 3^K T\left(\frac{n}{4^K}\right) + \sum \frac{3^i n^2}{16^i}$$

$$3^K T\left(\frac{n}{4^K}\right) + n^2 \sum \left(\frac{3}{16}\right)^i \quad K = \log_4 n$$

$$3^{\log_4 n} T(1) + n^2 \sum \left(\frac{3}{16}\right)^i$$

$$n^{\log_4 3} \cdot 1 + O(n^2) = \theta(n^2)$$

